

Aeronautical Systems B

Lecture: 3

1. Elements of drag
2. Lifting-line theory

Drag acting on the wing

■ D'Alembert's Paradox

No drag acts on an object of arbitrary shape placed in a uniform flow of a perfect fluid (inviscid and incompressible).

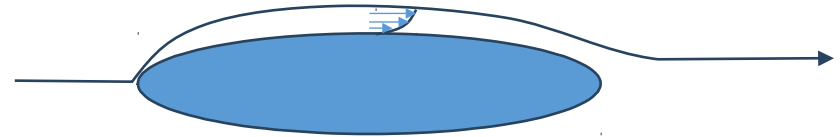
■ Actual drag

Drag = Induced Drag + Parasite Drag

Parasite Drag = Friction Drag + Profile Drag + other drag

Friction drag

In the laminar boundary layer, a shear stress proportional to the velocity gradient occurs (proportionality constant: viscosity). The stress on the wing surface is the frictional drag.

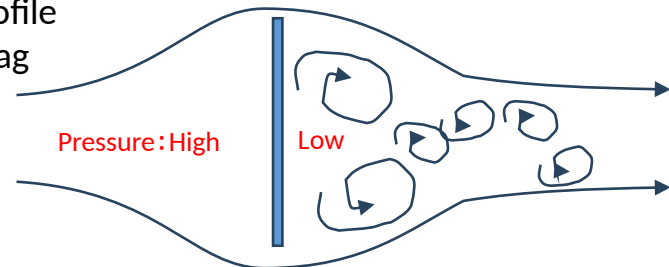


Induced Drag : Drag caused by lift

Friction Drag : Drag due to friction on the object's surface

Profile Drag: Drag due to pressure on the object's surface (Pressure drag)

Profile Drag

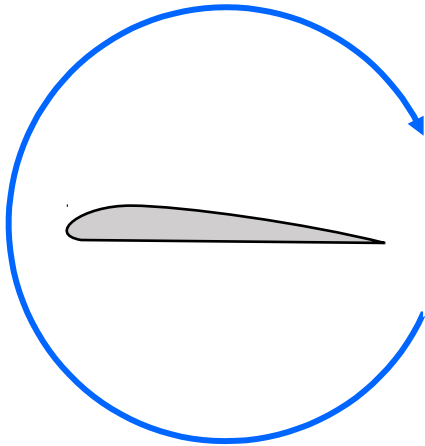


Drag acting on the wing

■ Induced Drag

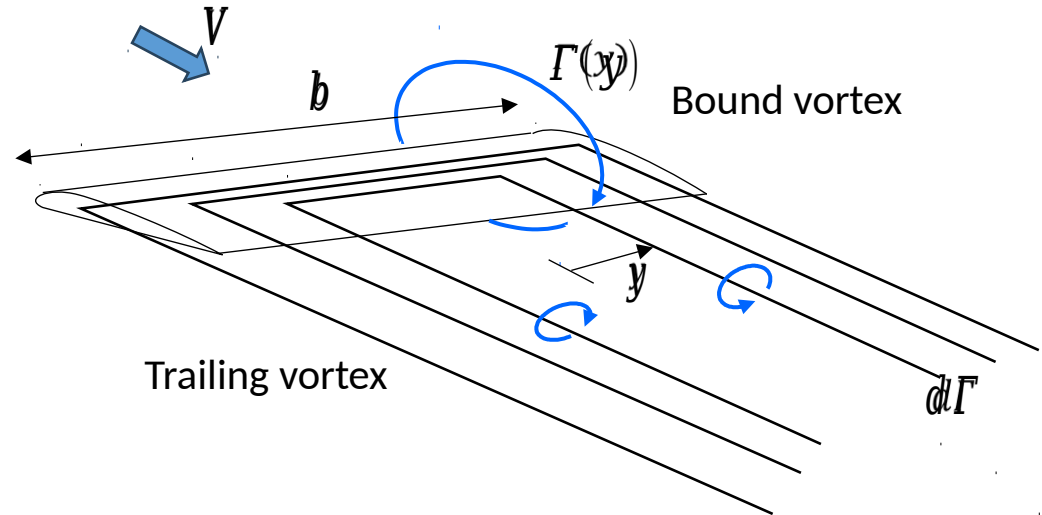
2D Wing

Circulation Γ due to vorticity



$$L = \rho V \Gamma$$

3D Wing



The trailing vortex induces an additional velocity on the wing. As a result, it reduces the angle of attack by the induced angle of attack α_i .

Drag acting on the wing

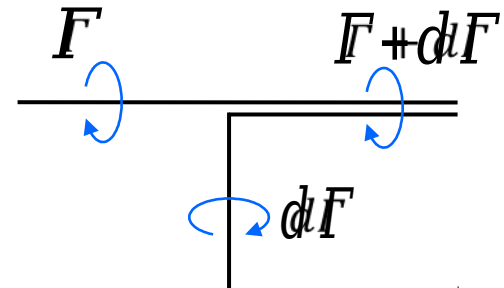
■ Induced Drag $\Gamma(y)$ vortex intensity, Density of circulation (vortex filament)

Relationship between vortex intensity $\Gamma(y)$ and lift distribution $dL(y)$ (lift generated by the airfoil element $y \sim y + dy$)

$$dL(y) = \rho V \Gamma(y) dy \quad (B4-1)$$

Relationship between the lift coefficient $C_l(y)$ and the lift distribution $dL(y)$

$$dL(y) = \frac{1}{2} \rho V^2 C_l(y) c(y) dy \quad (B4-2)$$



By setting these equal

$$\Gamma(y) = \frac{1}{2} C_l(y) c(y) V \quad (B4-3)$$

Where,

$$\Gamma(\frac{b}{2}) = 0 \quad (B4-4)$$

It is assumed that all bound vortices become trailing vortices flowing backward from the trailing edge.

Drag acting on the wing

The trailing vortex generates an induced velocity $w(y)$ along the lifting line.

$$C_{l_l} \equiv C_{l_{l_0}} (\alpha - \alpha_i) \quad (\text{B4-5})$$

$$\alpha_i = \frac{w}{V} \quad (\text{B4-6})$$

Induced velocity at y generated by $d\Gamma(y)$

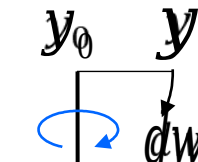
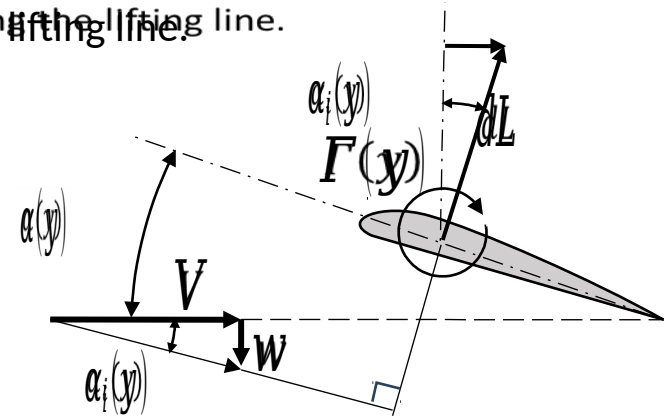
$$dw(y) \equiv \frac{d\Gamma(y)}{4\pi(y_0 - y)} \quad (\text{B4-7})$$

Integrate and divide by V

$$\alpha_i(y_0) = \frac{1}{4\pi V} \int_{-b/2}^{b/2} \frac{d\Gamma(y)}{y_0 - y} \quad (\text{B4-8})$$

Performing coordinate transformation $y = \frac{b}{2} \cos \theta$,

$$\alpha_i(\theta) = \frac{1}{2\pi bV} \int_0^\pi \frac{d\Gamma(\theta)}{\cos \theta_0 - \cos \theta} \quad (\text{B4-9})$$



Biot-Savart's Law

Infinite-length vortex line

$$dw(y) \equiv \frac{d\Gamma(y)}{2\pi(y_0 - y)} \quad (\text{B4-10})$$

Semi-infinite vortex line

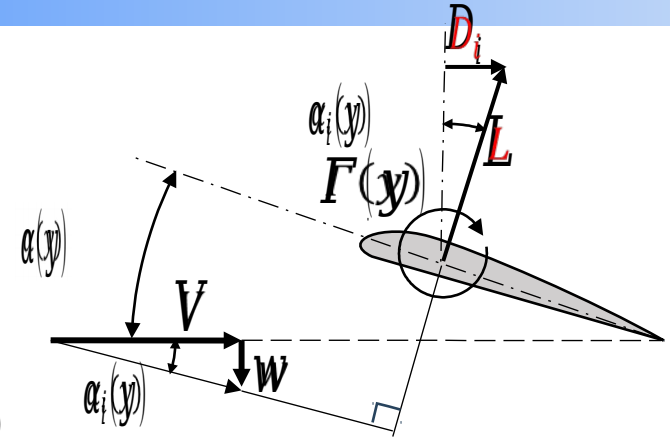
$$dw(y) \equiv \frac{d\Gamma(y)}{4\pi(y_0 - y)} \quad (\text{B4-11})$$

Drag acting on the wing

Assume an elliptical distribution

$$F \equiv F_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (\text{B4-12})$$

$$\alpha_i(\theta) = \frac{F_0}{2\pi b W} \int_0^\pi \frac{\cos\theta d\theta}{\cos\theta_0 - \cos\theta} = \frac{\Gamma_0}{2bW} \quad (\text{B4-13})$$



When the wing has no twist (α is constant), the section lift coefficient remains constant along the spanwise direction.

Lift

$$L = \frac{1}{2} \rho V^2 C_L \int_{-1/2}^{1/2} c dy = \frac{1}{2} \rho W^2 C_L S \quad (\text{B4-14})$$

$$C_L \equiv C_{L_i} \quad \text{Lift coefficient for an angle of attack reduced by } \alpha_i \quad (\text{B4-15})$$

$$C_{L_i}(y) \equiv \frac{2\Gamma_0}{C_{L_i} W} \sqrt{1 - \left(\frac{2y}{b}\right)^2} = C_{L_0} \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (\text{B4-16})$$

The wing planform is elliptical.

Drag acting on the wing

$$\alpha_i = \frac{F_\theta}{2\pi b V} = \frac{\frac{11}{22} \alpha_0 C_L}{2bV} = \frac{\alpha_0 C_L}{4b} = \frac{C_L}{\pi A} \quad (\text{B4-17})$$

Aspect ratio:

$$A = \frac{b^2}{S} = \frac{b^2}{\pi c_0 b/4} = \frac{4b}{\pi c_0} \quad (\text{B4-18})$$

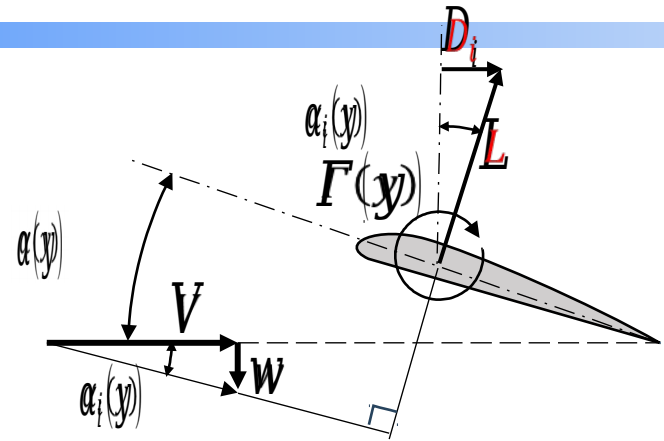
$$\alpha_i = \frac{C_L}{\pi A} \quad (\text{B4-19})$$

The induced angle of attack α_i is inversely proportional to the aspect ratio A.

$$C_L = C_{L_\alpha} \left(\alpha - \frac{C_L}{\pi A} \right) \quad (\text{B4-20})$$

Solving for C_L

$$C_L = C_{L_\alpha} \left[\frac{1}{1 + \left(\frac{C_{L_\alpha}}{\pi A} \right)} \right] \alpha \quad (\text{B4-21})$$



Induced Drag:

$$D_i = L \sin \alpha_i = L \alpha_i \quad (\text{B4-22})$$

$$C_{D_i} = \alpha_i C_L = \frac{C_L^2}{\pi A} \quad (\text{B4-23})$$

Substituting the theoretical value $C_{l_\alpha} = 2\pi$

$$C_{L_\alpha} \equiv C_{l_\alpha} \frac{A}{A+2} \alpha \quad (\text{B4-24})$$

$$C_{L_\alpha} \equiv C_{l_\alpha} \frac{A}{A+2} \quad (\text{B4-25})$$

This is a good approximation when the aspect ratio is large.

More accurate estimation formula:

$$C_{L_\alpha} \equiv C_{l_\alpha} \frac{A}{2 + \sqrt{4 + A^2}} \quad (\text{B4-26})$$

Drag acting on the wing

It is known that an elliptical wing minimizes induced drag coefficient for a given lift coefficient when the aspect ratio A is the same.

$$C_{D_i} \equiv \alpha_i C_L = \frac{C_L^2}{\pi A} \quad : \text{Induced drag of elliptical wings} \quad (\text{B4-23})$$

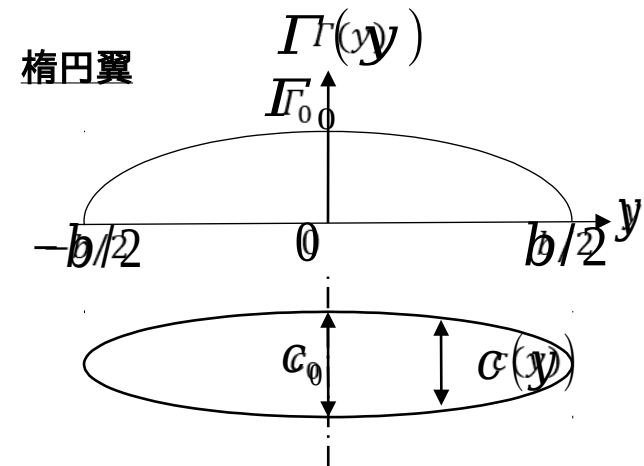
Generally

$$C_{D_i} \equiv \frac{C_L^2}{\pi A} (1 + \delta) \quad (\text{B4-27})$$

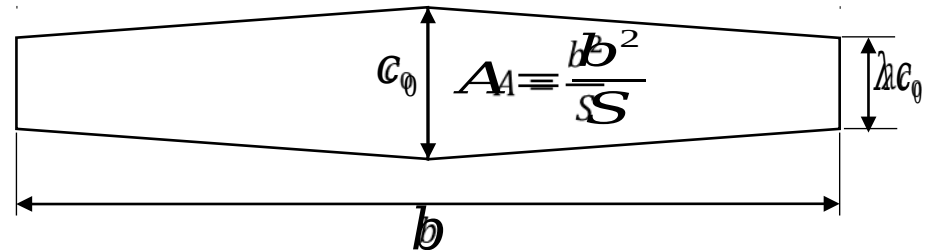
Example: Aspect ratio, $A = 6$, taper ratio $t = 0.5$, slope of lift curve is 5.5 (Theoretical value is 2π)

$$\delta = 0.0122$$

1.22% larger than the elliptical wing



Tapered wing



Drag acting on the wing

Drag = Induced Drag + Parasite Drag

$$C_{D_D} = C_{D_i} + C_{D_p} \quad (B4-28)$$

It is known empirically that parasite drag also has a component dependent on lift. To make it easier to handle, it is expressed as the square of the lift coefficient.

$$C_{D_p} = C_{D_{pmin}} + k C_L^2 \quad (B4-29)$$

Therefore, the total drag coefficient is

$$C_{D_D} = \frac{C_L^2}{\pi A} (1 + \delta) + C_{D_{pmin}} + k C_L^2 = C_{D_{pmin}} + \left(k + \frac{1 + \delta}{\pi A} \right) C_L^2 = C_{D_{pmin}} + \frac{C_L^2}{\pi e A} \quad (B4-30)$$

$$e = \frac{1}{1 + \delta + k \pi A} \quad \text{Oswald Efficiency Factor} \quad (B4-31)$$